



Power notation

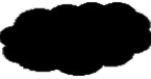
Task A: Notation

1) Jacob has spilt ink all over his maths notes. Identify what was under the spilt ink.



The base is 

The exponent is 

We say "12 

It means 12×12



The base is 

The exponent is 

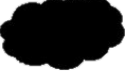
We say "9 to the power of 

It means
 $9 \times 9 \times 9 \times 9$



The base is 

The exponent is 

We say "4 

It is 64

2) Put these numbers in ascending order, smallest to largest.

9^2	2^3	9×2
10×4	10^4	$2 + 2 + 2$

3) These numbers have been put in ascending order. They are missing an **exponent**, a **base** or both.

$$2^{\square}, 5^{\square}, 3^{\square}, \square^2, \square^{\square}$$

a) Fill in the missing numbers using the following:

2	2	3	4	8	10
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b) Try and find a different way to complete part a)

Task B: Simplifying using power notation

1) Simplify the following using **exponents**:

a) $2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3$

b) $2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3$

c) $5 \times 2 \times 2 \times 5 \times 5 \times 5 \times 3 \times 2 \times 3 \times 3$

2) Pair each statement with the correct simplification.

Two squared multiplied by five to the power of three.

$$4^7 \times 7^4$$

Three squared multiplied by two to the power of five.

$$2^4 \times 7^4$$

Two squared multiplied by two squared multiplied by seven to the power of four.

$$2^5 \times 3^2$$

Four to the power of seven multiplied by seven to the power of four.

$$2^2 \times 5^3$$

3) Simplify the following using **exponents** and write the expression in words:

$$2 \times 2 \times 2 \times 2 \times 9 \times 9$$

$$2 \times 2 \times 3 \times 3 \times 5 \times 5$$

$$2 \times 3 \times 7 \times 2 \times 3 \times 7 \times 2$$